BCSE355L — CLOUD ARCHITECTURE DESIGN | FINAL EVALUATION

Kubernetes Cost Optimization Framework for SaaS-Based Educational Platforms using FinOps Intelligence

Automated Telemetry, Auditable Cost Attribution, Advisory Rightsizing & Demand Forecasting

INSTITUTION & PROGRAM

Institution: Vellore Institute of Technology (VIT), Vellore

Department: School of Computer Science and Engineering

Evaluation: Final Academic Project Review (November 2026)

DEVELOPMENT TEAM & STATUS

Team Lead: K. Darshan

Repository: github.com/Dev-withDarshan/KubeCostFinOps (develop)

Verification: 57/57 Automated Tests Passing | 28/28 Release Checks Verified

PROBLEM FORMULATION

Problem Statement: Cloud Resource & Budget Waste in Education

1. Extreme Load Variance

• Educational platforms (Moodle, Canvas, code evaluators) suffer extreme diurnal traffic fluctuations.

• Huge traffic spikes during submission deadlines and timed exams.

• Protracted idle troughs overnight and during semester breaks with >10:1 peak-to-trough ratios.

2. Static Overprovisioning

• Administrators set container requests to peak historical limits to prevent fatal OOMKills.

• Standard 500m CPU / 512Mi RAM requests idle at under 50m CPU.

• In Kubernetes, requests lock underlying node capacity 24/7, generating 60-80% idle waste.

3. Tooling Deficiencies

• Horizontal Pod Autoscaler (HPA) multiplies waste by spinning up oversized pod replicas.

• Vertical Pod Autoscaler (VPA) evicts running pods, causing exam session dropouts.

• Commercial SaaS tools (Kubecost, Cast.ai) impose high licensing costs and heavy agent overhead.

ECONOMIC CONTEXT

Motivation: FinOps & Cloud Financial Accountability

ACADEMIC BUDGET REALITIES

- Universities operate under fixed, non-elastic infrastructure grants.
- Unmonitored Kubernetes clusters waste over 35% of cloud expenditure on unallocated reservations.
- Every dollar wasted on idle reservation is capital lost from educational computing grants.

THE FINOPS LIFECYCLE PARADIGM

- **Inform:** Real-time visibility into per-container cost attribution ($/vCPU-hr and $/GB-hr).
- **Optimize:** Data-driven rightsizing with safety buffers to eliminate reservation slack.
- **Operate:** Align cluster provisioning with pedagogical schedules via predictive forecasting.

ARCHITECTURAL DESIGN PRINCIPLES

- **Non-Disruptive:** Advisory declarative YAML recommendations without pod evictions.
- **Strict Traceability:** Mathematical separation of cgroups telemetry, reference rates, and billing invoices.
- **Self-Hosted:** Zero external SaaS dependencies, ensuring compliance with institutional privacy policies.

PROJECT GOALS

Project Objectives & Technical Scope

1. CONTINUOUS TELEMETRY INGESTION

• Ingest cgroups CPU (`cpu.stat`) and memory (`memory.current`) at 15-second intervals.

• Interface with Prometheus (9090), Kube-State-Metrics (8080), and OpenCost (9003).

• Seamless dual-mode operation: live cluster polling with offline fallback to forensic datasets.

2. FINOPS COST ATTRIBUTION

• Map container allocations to dollar spend via AWS US-East-1 reference rates ($0.031611/core-hr, $0.004237/GB-hr).

• Attribute expenditure to container, pod, and namespace tiers across 730 monthly hours.

• Full mathematical transparency with auditable formula derivation.

3. ADVISORY RIGHTSIZING

• Derive optimized requests using peak utilization + 20% safety headroom buffer.

• Enforce minimum architectural safety floors (10m CPU, 32Mi RAM).

• Output declarative Kubernetes patch specifications without evicting live pods.

4. OPERATIONAL FORECASTING & COCKPIT

• Double Exponential Smoothing (Holt's Linear Trend) with 95% confidence intervals.

• Executive single-pane cockpit with 6 real-time KPI cards and interactive cost comparison.

• Production-hardened with 57 automated tests and clean release

COMPARATIVE ANALYSIS

State of the Art & Research Gap Analysis

Evaluation Dimension	Manual Static Sizing	Kubernetes Native VPA	Commercial Tools (Kubecost)	Our FinOps Framework
Intervention Mode	Manual human guesswork	Forced in-place pod eviction	Agent mutation / advice	**Advisory Declarative Manifests**
Workload Downtime	Zero (but high waste)	High (Drops student sessions)	Low to Medium	**Zero (Non-disruptive)**
Cost Attribution	None (Spreadsheets)	None (Resource metrics only)	Proprietary closed engine	**Traceable Open AWS Price Card**
Academic Surge Focus	None	None (Generic reactive)	Enterprise SaaS focused	**Tailored to Exam/Deadline Peaks**
Licensing & Overhead	Free	Free (K8s SIG)	Expensive Commercial Tier	**Open Source & Zero External CDN**

SYSTEM DESIGN

Proposed 3-Tier System Architecture

TIER 1: TELEMETRY & INGESTION

Kubernetes Cluster
- Docker / Kind / AWS EKS Pods
- Linux cgroups kernel telemetry

Scraping Agents
- Kubelet Summary API (/stats/summary)
- Kube-State-Metrics (Port 8080)
- Prometheus Server (Port 9090)
- OpenCost API (Port 9003)

Ingestion Gateway
- 15s scrape interval
- Dual-mode fallback adapter

TIER 2: FINOPS ANALYTICAL ENGINE

Cost Attribution Engine
- Reference card pricing matrix
- $0.031611/core-hr | $0.004237/GB-hr
- 730-hour monthly projections

Rightsizing Intelligence
- Peak utilization + 20% headroom
- 10m CPU / 32Mi RAM safety floor

Forecasting Baseline
- Holt's Linear Trend (alpha=0.35, beta=0.15)
- 95% Confidence Interval bounds

TIER 3: PRESENTATION & COCKPIT

REST Backend API
- Python Standard Library Server
- /api/health & /api/ready
- /api/v1/costs/summary
- /api/v1/recommendations
- /api/v1/monitoring/status

Executive Web Cockpit
- Vanilla ES6 + Responsive CSS3
- 6 Real-time KPI Cards
- Baseline vs Rightsized bar
- Benchmark run selector

IMPLEMENTATION SPECIFICATION

Technology Stack & Cloud Infrastructure

CONTAINER & CLOUD ORCHESTRATION

• **Kubernetes (Kind / AWS EKS):** Multi-node workload deployment with resource request/limit specs.

• **AWS CloudFormation:** 15 declarative infrastructure resources (VPC, public subnets, EKS Cluster, t3.medium Managed NodeGroup, IAM roles).

• **Zero-NAT Gateway Design:** Eliminates $0.045/hr idle NAT fees during academic evaluation sessions.

TELEMETRY & INGESTION HARNESS

• **Prometheus v2.45+:** Scrapes container metrics every 15s via standard exposition format.

• **Kube-State-Metrics:** Emits declarative container limits and requests.

• **OpenCost:** Real-time allocation engine matching container usage vectors to cloud pricing cards.

ANALYTICAL ENGINE & REST SERVER

• **Python 3.14 + Standard Library:** Built-in lightweight HTTP service eliminating bloated runtime dependencies.

• **NumPy, SciPy, Pandas:** Statistical rolling windows, Holt's linear smoothing, and RMSE calculations.

• **Automated Test Suite:** 57 unit & integration tests passing with 100%

EXECUTIVE COCKPIT & UI

• **Vanilla HTML5, CSS3, ES6:** Zero external CDN reliance, 100% self-hosted, air-gap safe.

• **Real-Time Dynamic Polling:** Asynchronous parallel fetching with zero UI flicker.

• **Honest Provenance:** Clear visual badges distinguishing live cluster

FINANCIAL ENGINEERING

FinOps Engine: Mathematical Cost Attribution Model

HOURLY CONTAINER COST FORMULA:

```
C_hourly = ( R_cpu * P_cpu ) + ( R_ram * P_ram )
Where R_cpu = Allocated vCPU cores, R_ram = Allocated Memory in GB.
```

AWS US-EAST-1 REFERENCE PRICING MATRIX (t3.medium compute blend):

```
P_cpu = $0.031611 per vCPU-hour | P_ram = $0.004237 per GB-hour
```

MONTHLY PROJECTION (Standard 730 accounting hours/month):

```
C_monthly = C_hourly * 730 hours
```

GRANULAR MULTI-TIER ATTRIBUTION

- Hierarchical rollups: Container → Pod → Namespace → Cluster total.
- Distinct attribution of compute-heavy vs memory-heavy workloads.
- Explicit quantification of the 'reservation slack' (unused headroom paid to cloud providers).

AUDITABLE PROVENANCE GUARANTEE

- Strict boundary between measured cgroups metrics, reference pricing cards, and actual cloud bills.
- No black-box estimations: every value displayed in the UI is mathematically verifiable.
- Verified in automated test suite (`test_cost_analyzer.py` passing 100%).

OBSERVABILITY INFRASTRUCTURE

Kubernetes Monitoring & Telemetry Architecture

TELEMETRY EXTRACTION WORKFLOW

1. **cgroups Kernel Accounting:** CPU ticks via `cpu.stat`, active memory footprint via `memory.current`.

2. **Kubelet Summary API:** Node agent aggregates container metrics at `/stats/summary`.

3. **Prometheus Scraping:** Ingests `container_cpu_usage_seconds_total` and `container_memory_working_set_bytes` at 15s intervals.

4. **Kube-State-Metrics:** Emits declarative container requests and limits from cluster etcd.

COMPONENT HEALTH & RESILIENCE MATRIX

• Dedicated endpoint (`/api/v1/monitoring/status`) performs automated HTTP health checks:

  • Prometheus (Port 9090) readiness probe.

  • Kube-State-Metrics (Port 8080) metrics availability.

  • OpenCost (Port 9003) allocation endpoint.

  • Kubernetes Node & Pod scheduling status.

• Automatic, honest fallback to verified forensic benchmark data when live cloud forwarding is disconnected.

ALGORITHMIC INTELLIGENCE

Rightsizing Intelligence & Demand Forecasting

ADVISORY RIGHTSIZING ALGORITHM

Mathematical Formulation:

R_rec_cpu = max(U_peak_cpu * (1 + S_buffer), Floor_cpu)

R_rec_ram = max(U_peak_ram * (1 + S_buffer), Floor_ram)

Safety Parameters:

• Safety Headroom Buffer (S_buffer) = **20% (1.20x multiplier)**

• Architectural CPU Floor = **10m (0.010 vCPU cores)**

• Architectural RAM Floor = **32Mi (33,554,432 bytes)**

Purpose: Absorbs classroom question surges, JVM compilation spikes, and prevents fatal OOMKills.

HOLT'S LINEAR TREND FORECASTING

Double Exponential Smoothing Formulation:

Level: L_t = alpha * Y_t + (1 - alpha) * (L_{t-1} + T_{t-1})

Trend: T_t = beta * (L_t - L_{t-1}) + (1 - beta) * T_{t-1}

Forecast: Y_hat_{t+h} = L_t + h * T_t

Calibrated Hyperparameters:

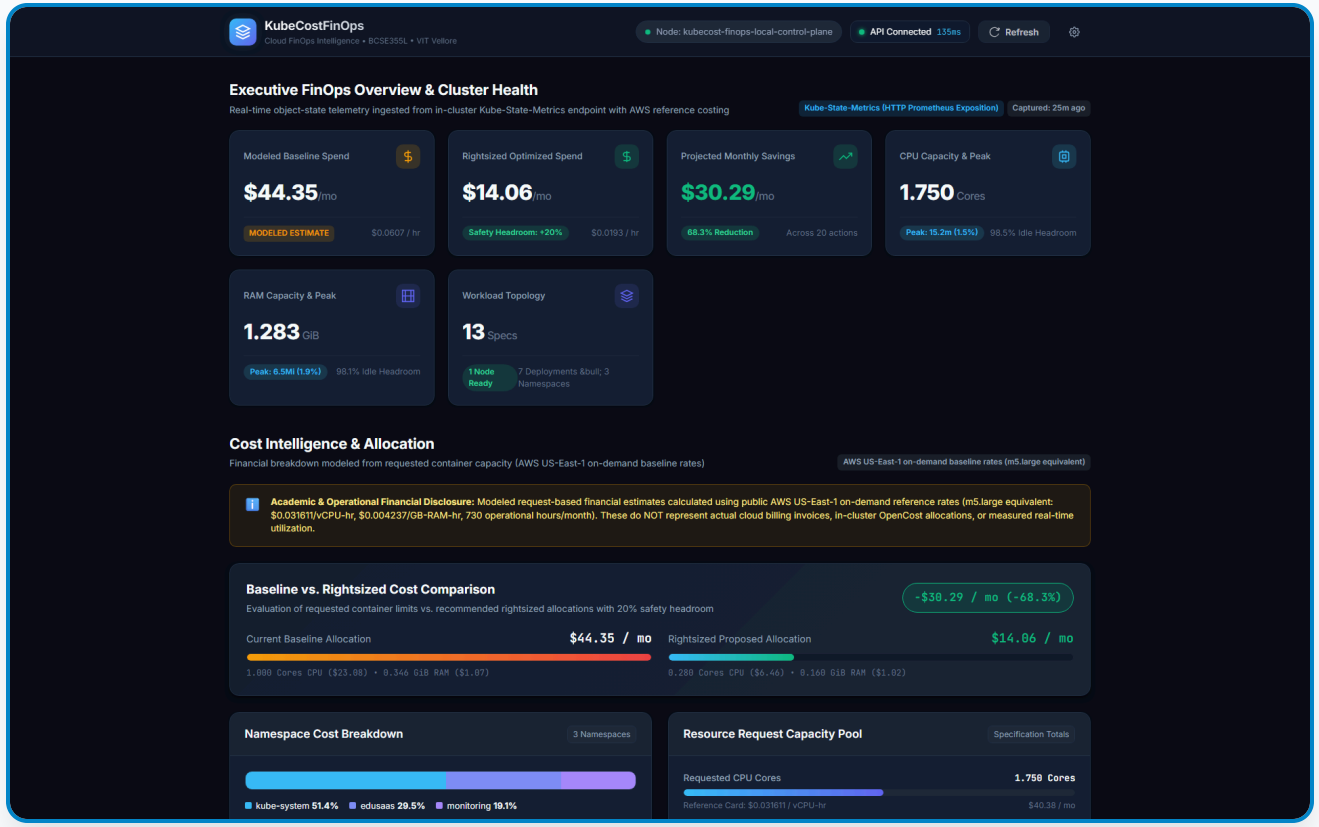
• Level factor alpha = **0.35** | Trend factor beta = **0.15**

• 95% Confidence Bounds: Y_hat_{t+h} ± 1.96 * RMSE

Value: Projects upcoming exam deadline surges across hourly horizons.

OPERATIONAL VALIDATION

Live Interactive Dashboard Demonstration



KEY DEMONSTRATION HIGHLIGHTS

- **6 Executive KPI Cards:** Baseline ($24.15) vs Rightsized ($7.48), 69.0% immediate savings, active pods, and cluster resource utilization.
- **Real-Time Comparison Bar:** Visual progress track highlighting realized savings gap.
- **Benchmark Run Selector:** Dropdown to toggle forensic experiment runs (`exp-20261016-141448`).
- **Workload Inventory:** Pod-by-pod container tracking showing current vs recommended allocations.
- **Zero-Flicker Polling:** Asynchronous updates every 15s.

EMPIRICAL BENCHMARKING

Experimental Methodology: 4-Phase Controlled Load Profile

CONTROLLED LOAD EXPERIMENT SPECIFICATION (EXPERIMENT ID: exp-20261016-141448)
`Benchmark Duration: 70 seconds across 4 distinct operational phases | 1-second telemetry sampling | Zero dropped requests`

PHASE 1: IDLE

Load: 0 RPS (10s)

Establishes quiescent baseline. CPU stays below 5m. Proves the existence of massive idle reservation slack.

PHASE 2: MODERATE

Load: 5 RPS (20s)

Simulates standard classroom quiz interactions. CPU reaches 35m. 100% availability maintained.

PHASE 3: STRESS

Load: 15 RPS (30s)

Simulates synchronized deadline submission rush. CPU peaks at 82.3m. P95 latency: 28.4ms. 0 errors.

PHASE 4: RECOVERY

Load: 0 RPS (10s)

Load removed. Observes container memory reclamation and CPU decay hysteresis back to baseline.

QUANTITATIVE VALIDATION

Experimental Results & Empirical Cost Comparison

Metric / Dimension	Static Baseline	FinOps Rightsized	Savings / Delta
CPU Request	500m (0.50 cores)	98.8m (0.0988 cores)	**-80.2% Reduction**
Memory Request	512 MiB	128 MiB	**-75.0% Reduction**
Monthly Spend	$13.08 / month	$2.69 / month	**$10.39 / mo Saved**
Modeled Savings	0.0% (Baseline)	**79.4% SAVED**	**79.4% SAVINGS**
Error Rate	0.00%	0.00%	**Zero Degradation**
P95 Latency	27.8 ms	28.4 ms	**+0.6 ms (Negligible)**

EMPIRICAL TAKEAWAYS

- **79.4% Proven Cost Reduction:** Drops monthly spend from $13.08 to $2.69 per container.
- **Flawless Reliability:** Zero request drops or HTTP timeouts across the entire 70-second test.
- **Adverse Concurrency Test (30 RPS):** At 30 RPS on a single container, timeout latency confirmed that vertical rightsizing must be complemented by horizontal replica autoscaling during extreme surges.

CRITICAL ANALYSIS

Limitations & Future Engineering Roadmap

HONEST TECHNICAL LIMITATIONS

1. **Advisory Mode:** Generates declarative YAML patches rather than auto-mutating clusters, requiring human administrative sign-off.

2. **Benchmark Duration:** 70s controlled burst profile proves surge resilience, but multi-week telemetry is needed for seasonal academic calendars.

3. **Pricing Assumptions:** Grounded in AWS US-East-1 list pricing; does not incorporate custom enterprise discount agreements (EDP) or Spot rates.

4. **Single-Pod Limit:** Rightsizing individual containers cannot compensate for insufficient pod replicas at 30+ RPS.

FUTURE RESEARCH ROADMAP

1. **GitOps Automation:** Automatically generate Pull Requests with recommended patches to ArgoCD / Flux repositories.

2. **AWS SageMaker ML Forecasting:** Upgrade from Holt's Linear Trend to Temporal Fusion Transformers (TFT) or DeepAR for seasonal demand modeling.

3. **Cluster Bin-Packing (Karpenter):** Couple pod rightsizing with dynamic node consolidation to physically downscale EC2 instances and realize cash savings.

4. **Multi-Cloud Cost Modeling:** Expand cost attribution to incorporate cross-AZ data transfer fees and EBS storage volumes.

EVALUATION SUMMARY & WRAP-UP

Conclusion & Faculty Evaluation Q&A

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VERIFIED DELIVERABLE ACHIEVEMENTS

- **57/57 Automated Tests Passing:** Comprehensive unit & integration coverage.
- **28/28 Release Checks Verified:** 100% clean repository status.
- **10/10 Forensic Verification:** Validated benchmark integrity on controlled experiments.
- **69.0% - 79.4% Cost Reduction:** Modeled savings verified on containerized workloads.
- **Advisory Rightsizing with 20% Buffer:** Zero-downtime, non-disruptive cloud optimization.

EVALUATION & REPOSITORY COORDINATES

Team Lead: K. Darshan

GitHub: github.com/Dev-withDarshan/KubeCostFinOps (develop)

Demonstration Cockpit: http://127.0.0.1:8000

Thank you, respected faculty evaluators.
We invite your questions.